

ONLINE RETAIL DATA ANALYSIS

Professional Data Analysis Report

Project: EncoderX Internship

Dataset: Online Retail Dataset

Analysis Tools: Python, Pandas, NumPy, Matplotlib, Seaborn

This report presents a structured analysis of an Online Retail transaction dataset. The project covers data cleaning, preprocessing, exploratory data analysis, visualization, key business insights, and recommendations.

1. Introduction

The objective of this project is to analyze online retail transaction data and identify important sales patterns, product performance, customer behavior, market concentration, and relationships between transaction variables. The analysis is intended to support data-driven business decisions related to products, customers, markets, inventory, and marketing.

2. Dataset Description

The original dataset contains 541,909 transaction records and 8 features: InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country. The dataset represents online retail transactions and includes both sales and cancellation-related records.

Feature	Description
InvoiceNo	Unique invoice/transaction identifier
StockCode	Product/item code
Description	Product description
Quantity	Number of units in a transaction
InvoiceDate	Transaction date and time
UnitPrice	Price per unit
CustomerID	Customer identifier
Country	Customer country

3. Data Cleaning and Preprocessing

- **Duplicate records:** 5,268 duplicate rows were identified and removed.
- **Missing values:** Missing Description values were removed. CustomerID had approximately 25.16% missing values and was retained because removing these records would significantly reduce the dataset.
- **Date conversion:** InvoiceDate was converted to datetime format.
- **Transaction inconsistencies:** Negative quantities were identified as cancellation/return-related records. For sales analysis, only Quantity greater than zero was retained.
- **Price validation:** Two records with negative UnitPrice were removed. Zero-price records were retained.
- **Outlier treatment:** IQR-based outlier detection was applied to Quantity and UnitPrice. Extreme values were treated using IQR capping.
- **Revenue calculation:** Revenue was calculated as Quantity × UnitPrice after preprocessing.

4. Exploratory Data Analysis

Metric	Result
Original records	541,909
Final sales records	525,460
Total revenue	£7,181,908.88

Total quantity sold	3,637,523
Unique products	3,925
Unique countries	38
Unique customers	4,339
Average revenue per record	£13.67
Average unit price	£2.87

5. Major Findings

- The United Kingdom was the dominant market, generating approximately £6.17 million in revenue.
- November 2011 recorded the highest monthly revenue at approximately £1.06 million.
- December 2011 recorded the lowest monthly revenue in the analysis; however, December is a partial month in this dataset, so the result should not be interpreted as a full-month comparison.
- REGENCY CAKESTAND 3 TIER was the highest-revenue product, generating approximately £93,034.80 after preprocessing.
- JUMBO BAG RED RETROSPOT was the highest-selling product by quantity, with 21,664 units.
- Quantity had the strongest positive correlation with Revenue at 0.523.
- UnitPrice and Revenue had a weaker positive correlation of 0.239.
- Quantity and UnitPrice had a moderate negative correlation of -0.376.
- Revenue was concentrated among a relatively small number of high-value customers.

6. Visualizations

Five professional visualizations were created to communicate the main findings: Top 10 Products by Revenue (Bar Chart), Monthly Revenue Trend (Line Chart), Revenue Distribution (Histogram), Correlation Heatmap, and Quantity vs Revenue (Scatter Plot). These visualizations help identify product leaders, seasonal patterns, distribution shape, variable relationships, and unusual transaction behavior.

7. Key Business Insights

- The UK market is the primary revenue driver and should remain a major focus for customer retention and marketing.
- A small number of products contribute substantially to revenue, indicating opportunities for focused promotion and inventory planning.
- November represents a strong sales period, suggesting that marketing and stock preparation should be intensified before peak demand.
- High-value customers provide significant revenue and can be targeted through loyalty and personalized offers.
- Quantity is more strongly associated with revenue than unit price in this processed dataset.

8. Recommendations

- Focus marketing and customer-retention strategies on the United Kingdom market.
- Promote high-revenue products such as REGENCY CAKESTAND 3 TIER while monitoring inventory availability.

- Prepare inventory and marketing campaigns before peak periods such as November.
- Develop loyalty programs and personalized promotions for high-value customers.
- Use quantity and unit price together when evaluating product pricing and promotional strategies.
- Monitor unusual quantities and prices regularly to distinguish genuine bulk purchases from data-quality issues.

9. Conclusion

The Online Retail analysis demonstrates how data cleaning, exploratory analysis, and visualization can transform raw transaction data into actionable business insights. After preprocessing, 525,460 sales records were analyzed, revealing strong UK market dominance, clear product-level revenue differences, a notable November sales peak, and meaningful relationships between quantity, price, and revenue. The results provide a practical foundation for marketing, inventory, pricing, and customer-retention decisions.

Important note: Correlation measures association and does not by itself establish causation.

10. Tools and Technologies

Python • Pandas • NumPy • Matplotlib • Seaborn • Jupyter Notebook • Excel dataset • Git/GitHub

11. Project Deliverables

- Jupyter Notebook: ecommerce_analysis.ipynb
- Cleaned dataset: cleaned_online_retail.csv
- Analysis CSV files containing monthly, product, country, customer, and correlation results
- Five analysis charts
- README.md documentation
- Professional analysis report PDF